

Today's Changing Instrumentation World

With the acceptance of computers and modular devices, virtual instrumentation has undoubtedly been universally embraced in test and measurement areas. But there are more changes happening in this field. Let us have a look at some of them very briefly

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Scientists, academia and especially engineering professionals seem to be driving virtual instrumentation. A major benefit of virtual instrumentation is accelerated project development (which in turn drives better control and design) and cost-effective solutions. However, virtual versus traditional instrumentation is a very subjective issue.

While virtual instruments provide flexibility through user-defined devices, traditional box instruments have fixed, vendor-defined functionality but are powerful. Virtual systems are modular and fast whereas box instruments are traditionally easy to use with all the knobs, buttons and screen displays that are now also available with touchscreen functionality.

Virtual instrumentation in box

Thinking out of the box is regarded as a positive trait but sometimes thinking inside the box could also set a trend. Test and measurement companies are using PXI (PXI stands for PCI eXtensions for Instrumentation) modules to create one-box testers for various wireless communication applications such as base station manufacturing test, power amp test, RF IC test and MIMO test. With advanced processors, easy-to-use software and improvements in technology, PXIs are becoming even more powerful and a better alternative.

A PXI system is typically used by manufacturers of cell phones and other cellular devices. It not only offers the manufacturers faster time-to-market by incorporating controllers, chassis and existing PXI modules but it can also



Ni's VirtualBench

be upgraded easily by appending new instruments to it. Although every vendor is selling PXI modules separately, companies are pushing bundled solutions as they feel these solutions best demonstrate and represent customer and vendor advantages of using PXI-inside approach.

PXI is an open, PC-based platform with test, measurement and control capabilities in a low-cost, modular package. You can mix and match different open standard PXI modules to incorporate the exact control I/O and measurement required in each system. Triggering features and integrated timing in PXI backplane provide precise inter-module synchronisation. Vendors can easily pre-configure systems to match a customer's requirements and applications.

PXI Express multiplexers are capable of switching up to 300V DC or AC voltages and 2A current. They furnish around 152 channels for scanning multiple points to a common bus in 1-wire, 2-wire, or 4-wire configurations. The SMX-3276 3U PXI Express multiplexer from VTI Instruments gives ATE system designers the ability to switch

input signals with up to 34MHz bandwidth, while reducing crosstalk levels to as low as -70dB at 100kHz.

Another advantage that PXI has is the power of integration, which allows you to easily integrate another vendor's product into your instrument.

On the downside, especially in case of high-volume applications, PXI adds significant cost as compared to a custom-integrated solution. Also, industry experts believe that there is a lack of ruggedised PXI products for extreme environments. Commercial PXI products are typically not designed for harsh environments. Therefore, depending on the product, some kind of tweaking may be needed to modify the instrument, which could also add to cost. Some of these single-box PXI test devices look like traditional test-bench instruments.

Many instruments in one

"Today, a typical workspace of a typical design/test engineer includes an oscilloscope, a digital multimeter, a power supply and a function generator. Each instrument includes its own separate display and knobs/buttons,"